Attention-Based MIL Baseline on CAMELYON16, CAMELYON17 and BRACS

ECE1512 Project A, Part B — Baseline Evaluation Summary

Syed Rayyan Albeez Ikramullah

Code: https://github.com/Rayz8839/ece1512-projectA-mil

***Abstract*—We train the supplied attention-based multiple-instance learning (ABMIL) model on frozen, pre-extracted patch features for three whole-slide-image (WSI) benchmarks and report its test behaviour. A single run (seed 42) reaches 0.934 AUC on CAMELYON16 (2 classes), 0.839 on CAMELYON17 (4 classes) and 0.804 on BRACS (7 classes). Accuracy and macro-F1 fall sharply as the label space grows, and the loss curves show strong overfitting on the two harder tasks. Replacing learned attention with uniform pooling lowers AUC on every dataset, indicating that learned attention is beneficial in this setup. We also place the numbers in the context of published ABMIL results, which are not directly comparable because features, splits, class definitions and the number of runs differ.**

I. MODEL STRUCTURE AND TRAINING PROCEDURE

Input. Each slide is a bag of patch embeddings of dimension 384 taken from the provided HDF5 files; slides contain on average about 12.5k (C16), 10.6k (C17) and 9.0k (BRACS) patches. Only slide-level labels are available and the feature extractor is frozen, so only the MIL head is trained. The head has three parts (Table I): an encoder (Linear 384→256 + ReLU, applied per patch); a gated-attention aggregator [1], $a_i = w^T[\tanh(Vh_i) \odot \mathrm{sigm}(Uh_i)]$ with 128 hidden units and a softmax over the unmasked patches of a slide, which produces one weighted slide embedding; and a one-layer linear classifier (256→number of classes). The head has 164,995 trainable parameters in the 2-class case (165,509 for C17 and 166,280 for BRACS).

Training. Cross-entropy loss, AdamW (learning rate 1e-4, weight decay 1e-5), batch size of one slide, 30 epochs, seed 42. After each epoch the model is evaluated on a validation split and scored by AUC + macro-F1; the checkpoint with the highest score (first one in case of ties) is restored and evaluated once on the test split. CAMELYON16 keeps its official test slides, CAMELYON17 uses centers ≥3 as test, and 10% of the remaining slides form the validation set; BRACS uses the supplied split. Test sets are small and strongly imbalanced (Table II).

Metrics. We report accuracy, macro-averaged F1, cross-entropy loss and AUC. AUC is computed from the softmax probabilities by the supplied classification_metrics helper; for the multi-class tasks (C17, BRACS) it is the macro-averaged one-versus-all AUC, the convention also used in the literature on multi-class BRACS [7].

TABLE I
PARAMETERS PER COMPONENT (2-CLASS MODEL)

Component	Layers	Parameters
Encoder	Linear(384,256) + ReLU	98,560
Aggregator	Gated attention (V, U, w), 128-d	65,921
Classifier	Linear(256, #classes)	514
Total		164,995

TABLE II
SLIDE COUNTS AND CLASS DISTRIBUTION (LABELS 0..K−1)

Dataset	Split	Slides	Class counts
C16 (K=2)	Train	242	0:142, 1:100
	Val	27	0:16, 1:11
	Test	129	0:80, 1:49
C17 (K=4)	Train	269	0:177, 1:18, 2:37, 3:37
	Val	30	0:20, 1:2, 2:4, 3:4
	Test	200	0:121, 1:16, 2:18, 3:45
BRACS (K=7)	Train	389	0:24, 1:119, 2:54, 3:24, 4:28, 5:40, 6:100
	Val	62	0:10, 1:10, 2:7, 3:6, 4:8, 5:9, 6:12
	Test	86	0:6, 1:16, 2:9, 3:11, 4:12, 5:12, 6:20

II. BASELINE PERFORMANCE

Table III summarises the test results of the selected checkpoints. Fig. 1 shows loss, accuracy and validation AUC/F1 per epoch and Fig. 2 the test confusion matrices.

TABLE III
TEST PERFORMANCE OF ABMIL (SINGLE RUN, SEED 42)

Dataset	Best val ep.	Acc (%)	AUC	Macro-F1	Loss
C16	3	88.37	0.9342	0.8668	0.2804
C17	22	78.50	0.8391	0.5323	0.7047
BRACS	6	39.53	0.8042	0.3494	1.4265

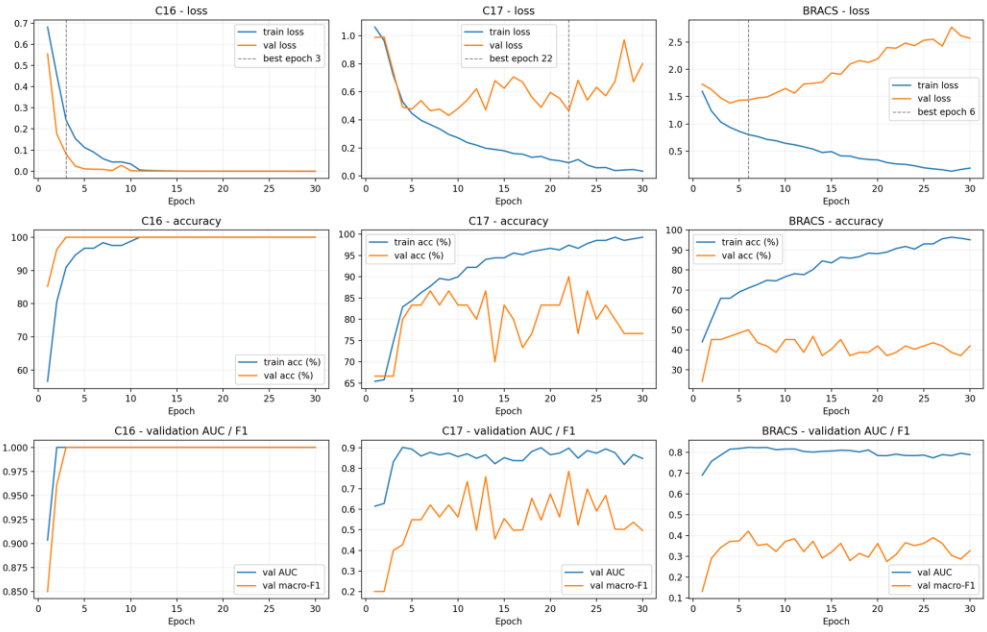


Fig. 1. Training/validation curves per dataset (columns: C16, C17, BRACS; rows: loss, accuracy, validation AUC/macro-F1). Dashed line: selected epoch (top row).

CAMELYON16. Training and validation loss approach zero within about 10 epochs, and validation AUC and F1 reach 1.0 from epoch 3 on. Because the validation set has only 27 slides, it cannot separate later epochs, so epoch 3 is kept. On the test set the model makes no false positives (precision 1.00 for class 1) but misses 15 of 49 positive slides (recall 0.69), which explains the 88.4% accuracy and the gap to the perfect validation score. Besides the small validation set, the official test slides may also differ in distribution from the training slides; we did not test this.

CAMELYON17. Training accuracy keeps rising to about 99%, whereas validation loss is lowest at epoch 9 and then becomes noisy and drifts upward; validation accuracy fluctuates between roughly 70% and 90% after the first epochs. The majority class and the largest abnormal class are handled well (F1 0.92 and 0.78), but classes 1 and 2 are weak (F1 0.24 and 0.19); 12 of the 18 class-2 test slides are predicted as class 3. The gap between accuracy (78.5%) and macro-F1 (0.53) reflects this imbalance. Note that the validation set contains only 2, 4 and 4 slides of classes 1–3, so checkpoint selection (epoch 22) is very noisy.

BRACS. This is the hardest task: validation loss is lowest at epoch 4 and then rises from about 1.4 to about 2.6 while training accuracy reaches about 96% and validation accuracy stays near 40%, a clear case of overfitting with 389 training slides and 7 classes. Only class 6 (F1 0.81) is recognised reliably; class 5 has the next-highest recall (0.58) but low precision (0.32, F1 0.41), and the remaining classes have F1 $\leq$ 0.33. Test accuracy is 39.5% and macro-F1 0.35, yet AUC is still 0.80, which means the ranking of class probabilities is informative even when the arg-max decision is wrong.

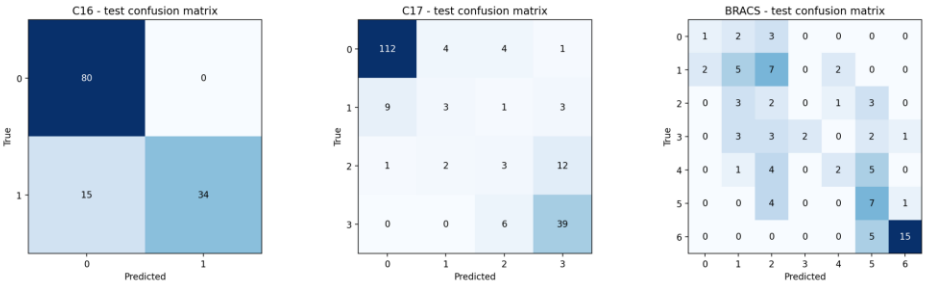


Fig. 2. Test confusion matrices (rows: true class, columns: predicted class).

III. CONTRIBUTION OF EACH COMPONENT

Encoder. It re-maps the generic frozen features into a smaller task-specific space (256-d) shared by the attention branch and the pooled embedding, and it holds 60% of the parameters. *Aggregator.* It converts a variable-size bag into one fixed-size vector by weighting patches by learned importance; this is the only place where evidence from different patches is combined. *Classifier.* A single linear layer maps the slide embedding to class logits, so any non-linearity must come from the encoder and the gating.

To measure the aggregator, we retrained each model with all attention logits set to zero (uniform mean pooling), keeping everything else fixed (Table IV). AUC drops by 0.28, 0.19 and 0.14 on C16, C17 and BRACS and macro-F1 by 0.28, 0.34 and 0.14, so learned attention is clearly beneficial in this setup. On C17 the mean-pooling model reaches 60.5% accuracy, which equals the share of class 0 in

the test set (121/200), and its macro-F1 of 0.19 matches predicting class 0 for every slide; its AUC of 0.644 shows that the probabilities still carry some ranking information, but its arg-max decisions are those of a majority-class predictor. This checkpoint was selected at epoch 1 on a 30-slide validation set.

TABLE IV
AGGREGATOR ABLATION (SINGLE RUN, SEED 42)

Dataset	Metric	Attention	Mean pool	Δ
C16	AUC	0.9342	0.6584	+0.2758
	Macro-F1	0.8668	0.5883	+0.2785
	Acc (%)	88.37	67.44	+20.93
C17	AUC	0.8391	0.6443	+0.1948
	Macro-F1	0.5323	0.1885	+0.3439
	Acc (%)	78.50	60.50	+18.00
BRACS	AUC	0.8042	0.6616	+0.1426
	Macro-F1	0.3494	0.2103	+0.1391
	Acc (%)	39.53	33.72	+5.81

Two caveats apply to this ablation. First, the encoder and classifier were not ablated, so the experiment shows that learned attention helps but does not rank the three components by importance. Second, the mean-pooling model keeps the (now unused) attention weights and is trained with the same learning rate and number of epochs as the attention model; a different training budget might narrow the gap.

Attention statistics on the test slides give a complementary view. The average normalised attention entropy (1 = uniform) is 0.84 (C16), 0.72 (C17) and 0.74 (BRACS), so attention is only mildly concentrated by this measure on C16 and moderately concentrated on C17 and BRACS. The top 1% of patches receive on average 26%, 38% and 46% of the attention mass, i.e. roughly 26–46 times their uniform share.

IV. CONTEXT: PUBLISHED ABMIL RESULTS

Table V places our numbers next to published ABMIL results. These are *not* directly comparable with ours: the papers differ in feature extractors, splits, class definitions and metric conventions, and they typically report means over several runs, whereas we report one seed. Published ABMIL on CAMELYON16 reaches 81.9–86.8% accuracy and 0.78–0.88 AUC [2]–[5]. Our C16 AUC of 0.934 lies above this range, which is plausible given that our frozen features come from a pathology-specific self-supervised encoder, although we did not test this explanation. For our CAMELYON17 task (4-class pN-stage, center-based split) and for 7-class BRACS we found no published ABMIL results under the same protocol: CAMELYON17 is mostly used as a binary task [8], and ABMIL on BRACS is usually reported for the 3-class task [2], [6]. We therefore do not claim that our baseline matches or exceeds published ABMIL.

TABLE V
OUR SINGLE-RUN ABMIL VS. PUBLISHED ABMIL

Dataset	This work	Published ABMIL (other protocols)
C16 (2-class)	Acc 88.4%, AUC 0.934	Acc 81.9–86.8%, AUC 0.78–0.88 [2]–[5]
C17 (4-class)	Acc 78.5%, AUC 0.839	No equivalent found; usually binary [8]
BRACS (7-class)	Acc 39.5%, AUC 0.804	No 7-class result found; 3-class: Acc 50.6%, AUC 0.724 [2] (not comparable)

V. CAVEATS

All numbers come from one seed and small test sets with few slides per minority class, so differences of a few points (especially macro-F1 on C17 and BRACS) are not statistically reliable; multiple seeds would be needed for mean ± std. The validation sets are also small (27, 30 and 62 slides, with as few as two slides for some classes), which makes checkpoint selection noisy. Attention weights show where the model looks, not causal evidence for the prediction. Any extension must reuse the same features, splits and seed to be comparable with these baselines.

APPENDIX A. REPRODUCIBILITY CHECKLIST (REPORT)

This checklist covers the report only. The code repository is a separate artefact and carries its own checklist in its README. "Where / details" points to the section of this report that answers each item, or gives the answer directly.

TABLE VI
MACHINE LEARNING REPRODUCIBILITY CHECKLIST (REPORT)

Item	Resp.	Where / details
Model and algorithm		
Clear description of the model	Yes	Sec. I; gated-attention equation; Table I.
Assumptions stated	Yes	Frozen pre-extracted features; slide-level labels only (Sec. I); single seed (Sec. V).
Complexity analysis	Yes	Trainable parameters per component (Table I). Mean training time, 30 epochs, Tesla T4: attention 222 s (C16), 268 s (C17), 393 s (BRACS); mean pooling 221 s, 263 s, 380 s.
Theoretical claims, with proofs	N/A	No theoretical claims are made.
Datasets		
Dataset statistics	Yes	Table II; mean patches per slide (Sec. I).
Train/validation/test splits	Yes	Sec. I and Table II. C16: official test slides; C17: centers >= 3 test, 10% of the rest validation; BRACS: supplied split.
Data excluded / preprocessing	N/A	Supplied 384-d HDF5 patch features are used unchanged; no slides were excluded.
Link to downloadable data	Yes	Zenodo record 23130543; Hugging Face dataset LeonYe/ECE1512-WSI-Feature (feature/{C16,C17,BRACS}/patch feats_pretrain_medical_ssl.h5).
Experimental results		

Item	Resp.	Where / details
All hyper-parameters specified	Yes	Sec. I: AdamW, lr 1e-4, weight decay 1e-5, batch size 1 slide, 30 epochs, D_inner 256, D_attn 128, no dropout. Learning rate is constant (the warmup_epoch key in the supplied YAML is not used by our training loop).
Hyper-parameter ranges and selection method	Yes	No tuning was done; the supplied configs were used. The checkpoint with the highest validation AUC + macro-F1 is selected (Sec. I).
Exact number of runs	Yes	One training run per dataset and pooling variant (seed 42).
Metric definitions	Yes	Sec. I: accuracy, macro-F1, macro one-vs-rest AUROC on softmax probabilities, cross-entropy loss.
Seeds	Yes	Seed 42 for initialisation, data shuffling and (C16, C17) the validation split.
Measure of variation (std, error bars)	No	Single run per setting; stated as a limitation in Sec. V.
Statistical significance tests	No	Not applicable to single runs; differences of a few points are not claimed to be significant.
Computing infrastructure	Yes	Google Colab, one Tesla T4 GPU, PyTorch 2.11.0+cu130.
Code available	Yes	GitHub link on the title page; the repository has its own checklist in its README.

REFERENCES

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